

Equation (NTC)

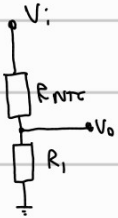
$$R(T) = R_0 \exp\left(\beta\left(\frac{1}{T} + \frac{1}{T_0}\right)\right)$$

R_0 = resistance at T_0

β is from data sheet, sometimes different betas are provided

for different measurement temperature range

Voltage divider



$$V_o = V_i \left(\frac{R_1}{R_{NTC} + R_1} \right)$$

Example Values used in calculations

Case	Temp °C	NTC Ω	desired V_o
1	25	10k	0V
2	85	1.5k	V_i

- Case 1 wants small R , Case 2 wants large R

$\therefore$ use geometric mean for R_1 : $R_1 = \sqrt{R_{NTCmax} \times R_{NTCmin}}$

- Voltage divider makes V_o approximately linear (compresses the exponential effect well enough within a small range)

Geometric Mean Proof

$$\begin{aligned} \text{Dynamic range} &= \left(\frac{R_1}{1.5 + R_1} \right) - \left(\frac{R_1}{10 + R_1} \right) \\ &= \frac{R_1(R_1 + 10) - R_1(R_1 + 1.5)}{(R_1 + 1.5)(R_1 + 10)} \\ &= \frac{8.5R_1}{R_1^2 + 11.5R_1 + 15} \\ a &= \frac{d}{dR_1} \left(\frac{8.5R_1}{R_1^2 + 11.5R_1 + 15} \right) = \frac{d}{dR_1} \left(\frac{f(R_1)}{g(R_1)} \right) \end{aligned}$$

$$f' = 8.5, \quad g' = 2R_1 + 11.5$$

$$a = \frac{8.5(R_1^2 + 11.5R_1 + 15) - 8.5R_1(2R_1 + 11.5)}{(R_1^2 + 11.5R_1 + 15)^2}$$

$$= \frac{-8.5R_1^2 + 127.5}{(R_1^2 + 11.5R_1 + 15)^2}$$

Set $a=0$ to find max dynamic range

$$8.5R_1^2 = 127.5$$

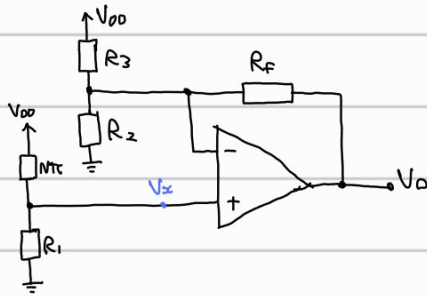
$$R_1 = \sqrt{\frac{127.5}{8.5}}$$

$$R_1 = \sqrt{10 \times 1.5}$$

$$127.5 = (10 - 1.5)(10 \times 1.5), \quad 8.5 = (10 - 1.5)$$

geometric mean of min/max NTC resistance

Sense circuit



$$V_+ = V_{DD} \left(\frac{R_1}{R_1 + R_{NTC}} \right)$$

$$V_- : \frac{V_{DD} - V_-}{R_3} - \frac{V_-}{R_2} - \frac{V_- - V_0}{R_f} = 0$$

$$-V_- \left(\frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_f} \right) + \frac{V_{DD}}{R_3} + \frac{V_0}{R_f} = 0$$

$$V_- = \frac{V_{DD} R_f + V_0 R_3}{R_3 R_f \left(\frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_f} \right)}$$

$\frac{V_{DD} R_f + V_0 R_3}{R_3 R_f}$

$$V_+ = V_- = V_x :$$

$$V_{DD} \left(\frac{R_1}{R_1 + R_{NTC}} \right) = \frac{V_{DD} R_f + V_0 R_3}{R_3 R_f \left(\frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_f} \right)}$$

$$V_{DD} \left(\frac{R_1}{R_1 + R_{NTC}} \right) \left(\frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_f} \right) R_f - V_{DD} \frac{R_f}{R_3} = V_0$$

$$V_x = \frac{V_{DD} R_f + V_0 R_3}{R_3 R_f \left(\frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_f} \right)}$$

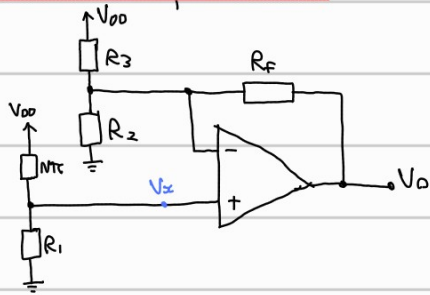
$$V_x R_3 R_f \left(\frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_f} \right) - V_{DD} R_f = V_0 R_3$$

$$V_0 = V_x \left(\underbrace{\frac{R_f}{R_3} + \frac{R_f}{R_2} + 1}_{\text{Gain}} \right) - \underbrace{V_{DD} \frac{R_f}{R_3}}_{\text{Offset}}$$

We can see this circuit will amplify and provide an offset to the NTC voltage divider output, we can use this to achieve our ideal output range

Full example

(for ENEL300 project)



$$V_o = V_x \underbrace{\left(\frac{R_F}{R_3} + \frac{R_F}{R_2} + 1 \right)}_{\text{Gain}} - V_{DD} \underbrace{\frac{R_F}{R_3}}_{\text{Offset}}$$

$$NTC = 10k @ 25^\circ C, 1.45k @ 85^\circ C$$

$$\therefore R_1 = \sqrt{10k \times 1.45k} = 3.81k \text{ for max dynamic range}$$

$$\text{choose } R_1 = 3.9k \text{ (E series)}$$

$$V_x = 3.3 \frac{R_1}{R_1 + R_{NTC}}$$

$$V_{x_{max}} = 3.3 \frac{3.9}{3.9 + 1.45} = 2.406V \quad \left. \vphantom{V_{x_{max}}} \right\} \rightarrow \text{Gain} = \frac{3.3}{2.406 - 0.926} = 2.23$$

$$V_{x_{min}} = 3.3 \frac{3.9}{3.9 + 10} = 0.926V$$

$$\text{Want DV at } 25^\circ C \therefore V_{x_{min}} \times \text{Gain} - \text{offset} = 0$$

$$\text{offset} = 0.926 \times 2.23 = 2.09 = V_{DD} \frac{R_F}{R_3}$$

$$\text{Let } R_F = 1k \Rightarrow R_3 = 3.3 \times \frac{1000}{2.09} = 1579 \Omega$$

$$\text{choose } R_3 = 1.58k \text{ (E series)}$$

$$\text{Gain} = \frac{R_F}{R_3} + \frac{R_F}{R_2} + 1$$

$$2.23 = \frac{1}{1580} + 1 + \frac{1}{R_2}$$

$$R_2 = \frac{1000}{0.5971} = 1674.8 \Omega$$

$$\text{choose } R_2 = 1.69k \text{ (E series)}$$

This gets us some initial values, but they aren't ideal because we rounded

V_x	V_o	Wanted
0.926	-0.0286	0
2.406	3.26	3.3

offset needs to be increased, so try $R_3 = 1.69k$

V_x	V_o	Wanted
0.926	0.069	0
2.406	3.30	3.3

This has slightly better dynamic range and reduces the BOM

Final $R_1 = 3.9k, R_2 = 1.69k, R_3 = 1.69k, R_F = 1k$